

iQSM_Plus — Open Recon Application Documentation

Version: 1.0.0 **Vendor:** HMRI Imaging Centre **Content qualification:**
RESEARCH — Not for diagnostic use

1. Purpose

iQSM_Plus is an Open Recon “image to image” application that reconstructs Quantitative Susceptibility Mapping (QSM) images from multi-echo gradient echo (GRE) magnitude and phase images, using the iQSM+ deep-learning pipeline (https://github.com/sunhongfu/iQSM_Plus).

The application receives magnitude and phase images from the standard product reconstruction pipeline (after distortion correction), performs QSM reconstruction (phase unwrapping, background field removal, and dipole inversion via a trained neural network), and returns a susceptibility map (in parts per million, ppm) as a new image series.

2. Input requirements

- **Sequence type:** 3D multi-echo gradient echo (GRE), monopolar readout
- **Required image output:** both magnitude and phase images must be enabled for the sequence
- **Recommended:** 3–8 echoes, TE spacing sufficient to avoid phase wraps between echoes at the field strength used
- **Prio Recon** must be disabled for the protocol (standard Open Recon requirement)

If phase images are not enabled, or magnitude/phase counts do not match, reconstruction will fail and an error will be logged; no images will be returned in that case.

3. Output

- A single new image series containing the reconstructed QSM map (one image per slice), with pixel values in ppm (parts per million), typically in the range -0.5 to +0.5 ppm for brain tissue.
- Original magnitude/phase images from the standard reconstruction are not modified or duplicated by this application.

4. Processing overview

1. Magnitude and phase images are buffered until the full acquisition (all slices, all echoes) has been received.
2. Phase images are converted to radians.
3. Acquisition parameters (echo times, voxel size, B0 field strength and direction) are read from the MRD image headers.
4. The iQSM+ neural network performs phase unwrapping, background field removal, and dipole inversion in a single inference pass.
5. The resulting 3D susceptibility map is re-sliced into individual 2D images and returned.

Reconstruction time is approximately 2-5 minutes for a typical whole-brain acquisition, depending on available GPU compute.

5. Configurable parameters

Parameter	Description	Default
config	Recon module selected by the MRD server	qsm
customconfig	Override for config, not used in normal operation	(empty)

6. Safety information

This application is intended for research use only and is not intended for diagnostic use. All images produced are automatically labeled “not for diagnostic use” per Open Recon requirements for research applications. No masking of extra-cranial tissue is performed by default; background-field artifacts may be visible outside the brain in the returned images.

This application does not modify raw k-space acquisition, gradient waveforms, RF pulses, or any other aspect of the underlying pulse sequence. It operates only on already-reconstructed magnitude/phase image data, after the standard product reconstruction pipeline.

7. Known limitations

- Requires both magnitude and phase image output from the source sequence.
- No brain-extraction/masking step is applied by default in this version; users should expect background-field noise outside the brain parenchyma.
- Reconstruction requires the complete acquisition (all slices/echoes) to be received before any output is returned — no partial/incremental results are available during acquisition.

8. Contact

HMRI Imaging Centre — for support regarding this Open Recon application.